Light is Everywhere

Pupil sheet

Name: _____ **Date:** _____

Safety: Never look at the sun and never shine the torch into anyone's eyes. Walk with the box; do not run.

1. What I see on the screen

On the screen you see the **Illuminance** card (the light, in lux — written `lx`), next to it a bar with four landmarks — *night*, *classroom*, *cloudy day*, *sun* — and the chart underneath. You start with **Start**; you move on with the green button at the bottom right, you go back with **Back**, and the last green button says **Finish**. In every place you press **Capture twice**, a few seconds apart.

The sensor answers at once — there is nothing to wait for. Hold the box still for about **ten seconds**, until the line settles, then write the number rounded: "about four hundred", not the last trembling digit.

Two rules. No fingers on the sensor's window. And every time **write down which way you pointed it** — up, at the glass, or at the floor — because in the same place the direction changes the number.

2. What I do on my own

First I predict. I do not change my prediction after I measure.

1. Where is there more light? ☐ under the desk ☐ beside the window
And how many times more? I think about _____ times.

because _____

_____

2. Put an arrow on the line where you think our room is today:

below 1 lx	300–500 lx	1000–10 000 lx	50 000 lx
night	classroom	cloudy day	sun

Before I measure, I draw a map of the room. One dot for each place I am going to measure, with its name beside it. After I measure, I write the number next to each dot: I colour the bright place lighter and the dark place darker.

Now I measure. I put the box down, I point it, I step back, I wait ten seconds, then I write two readings. If the number is too big for me to write, I just circle the word in the last column.

Place	Reading 1 (lx)	Reading 2 (lx)	Pointed at	Circle one
Under the desk				dark / medium / bright
Beside the window				dark / medium / bright
Room centre				dark / medium / bright

A shadow, measured. I pass my hand over the sensor: from _____ lx it dropped to _____ lx. **Weather:** sunny / cloudy / rainy

How sure are we? The two readings under the desk differ by _____ lx. Is the jump from the desk to the window much bigger than that? ☐ yes ☐ no

Working it out. The biggest: _____ lx the smallest: _____ lx
The difference: _____ − _____ = _____ lx
How many times more *(this row is completed from year III upwards)*: _____ : _____ = _____ times

3. The conclusion I write

I predicted that _____

_____

The data show that _____

_____

The difference is explained by _____

_____

Write one sentence: why less light reaches under the desk. Use the word *blocks* or *shadow*.

_____

Exit ticket

Write one thing that surprised you today.

_____